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An Intelligent IoT Fuzzy Based Approach for Automated Indoor Air Quality Monitoring

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***Abstract*— Indoor air quality has a critical impact on human and animal health and well-being. Because of the confinement during the Covid-19 health crisis, millions of people were forced to stay at home. Therefore, living environments should incorporate an automated monitoring system to detect unfavorable air quality scenarios. Also, Internet of things (IoT) due to their ability to carry air low-cost sensing units have become a cheap alternative to monitor pollution values in a given area. However, enough still needs to be learned as these types of studies are expected to increase in the future years. In this paper, we have developed an intelligent IoT-fuzzy-based device based on low-cost sensors for indoor air quality monitoring (IoT-fuzzy-based AQM). It used a type-1 fuzzy inference system to improve the measurement accuracy of the low-cost sensors and to estimate the air quality index. Experiment tests are made in an indoor environment (home), where the obtained results have proved the effectiveness of the realized system.**

Keywords– IoT, Air quality monitoring, type-1 fuzzy inference system, low-cost sensors, calibration, Smart Home, Smart Risk, Smart City

I. Introduction

Inadequate indoor air quality is a real-world public health problem. Carbon monoxide (CO) poisoning is one of the main causes of accidental poisoning in indoor environments (home, closed workplace, school classrooms). Annually, according to [1] the worldwide cumulative mortality of CO poisoning are estimated at 35420 deaths. Although the worldwide incidence of CO poisoning has remained stable over the last 25 years [1], many countries have recorded increases in the number of CO poisoning victims over the last three years, for example, Algeria [2], Japan [3], the United States [4] and the United Kingdom [5]. The other most infected categories of pollutants on the health are the particulate matter **PM**. Waste incineration facilities, dust, fires, vehicles are a major source of PM. The particle size ranges between 2.5 (μm) (**PM2.5**) and 10 (μm) (**PM10**). For these reasons, we present in this paper an intelligent monitoring solution for indoor air quality. The main core of this solution is the use of the Internet of Things (IoT) and type-1 fuzzy sets.

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A. Previous works

Firstly, there are several previous works [6], [7], [8], [9], [10] and [11] that have addressed the problem of the high cost of existing commercial solutions, where low-cost IoT-based AQM architectures are developed based on the minicomputer NodeMcu (It's an Arduino micro-motherboard equipped with an ESP32 wifi communication module) and based on the Raspberry pi micro motherboard. The low-cost IoT-based AQM solutions have shown their effectiveness in real-world applications like the one tested in Hanyang University of Korea [12], in subway tunnels [13], and in hospitals [14].

Secondly, in recent years, an increasing number of researchers have explored the use of artificial intelligent techniques (AI) in IoT-based AQM systems, for example, the use of fuzzy controllers [15] [16] and the use of adaptive neuro-fuzzy inference system (ANFIS) model [17]. According to the obtained results, AI techniques have reduced power consumption on one hand, and on the other hand, they are useful to better manage and understand the air quality in an indoor area. Finally, Kalman Filter (KF) algorithm was used to improve the accuracy of the air quality monitoring IoT-based system [18].

B. The work contributions

Besides their effectiveness for enhanced living environments in smart cities, the IoT-based AQM solutions have some limitations [9], [10] can be summarized as follows: 1) The limitations regarding the processing capabilities; 2) The sensor's accuracy; 3) The incoherence that can occur between what is measured and the true pollution quantity in the case of a faulty sensor; 4) The absence of a systematic and open-source solution that can be used on a large scale for the management and monitoring of pollutant data stored in Cloud systems. Also, the communication technologies used, and energy consumption. Hence, in this paper we present a novel indoor AQM solution named IoT-fuzzy-based AQM system, where the main contributions of this work are summarized as follows: 1) An experiment method to calibrate the low-cost MQ-7 sensors; 2) A new approach based on type-1 fuzzy sets and two MQ-7 sensors to estimate the air quality index (AQI). The particularity of this solution compared to the existing solutions is that with this: the incoherence that can occur between what is measured and the true pollutant concentration in the case of a faulty sensor can be avoided. Moreover, it enhances the accuracy of the low-cost sensors; 3) A IoT-AQM algorithm is proposed; 4) We have used Google Sheets to build a web application to simplify the

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management and the analysis of the monitoring pollutants stored data on a large scale for Smart home and Smart factory.
In the next section, we present the IoT-fuzzy-based AQM system. Section III presents results and discussion. Finally, we point out some conclusions and possible improvements to our work.

II. The IoT-fuzzy-based AQM system

In this section, we present the IoT-fuzzy-based AQM system. Figure 1 presents a picture of the realized device. The objective of this system is to inspect, monitor, and manage indoor air quality automatically.

Fig. 1: A picture of the realized IoT-fuzzy-based AQM system.

The IoT-fuzzy-based AQM device is a two-tier systems:

- The electronic air quality measurement system responsible for data acquisition and monitoring;
- a type-1 Fuzzy inference system.

A. The electronic air quality measurement system

The electronic AQM system contains the D-duino micro motherboard (based on NodeMCU board, ESP8266 wifi module), two MQ-7 CO sensors, a Shenyei PPD42N fine matter (PM) sensor, a DHT22 temperature, and humidity sensor, an alarm module, and a real-time clock as presented in Figure 2. Otherwise, the D-duino micro motherboard has only one analog input which causes a problem to read from the two MQ-7 sensors. To overcome this problem we used an analog-digital converter with a serial I2C-bus interface, where we have selected the module named PCF8591.

B. Calibration of the MQ-7 gas sensor

The calibration of the MQ-7 sensor is an important procedure because it has a low accuracy. According his Datasheet, the sensor must go through high and low heating cycles to obtain correct measurements. During the low-temperature phase, CO is absorbed on the MQ-7 plate, producing significant data. During the high-temperature phase, the absorbed

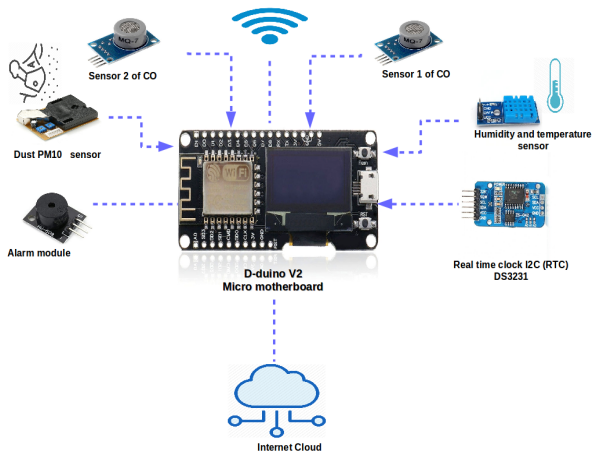

Fig. 2: electronic schematic of the IoT-fuzzy-besed AQM device

CO and other compounds (CO2) evaporate from the sensor plate, cleaning it for the next measurement. To carry out this procedure, a program under the Arduino board is required. The procedure of reading a CO concentration can be summarized as follows:

1) Apply 5 Volts for 60 seconds (heating step) ;
2) Apply 1.4 Volts (as Pulse Width Modulation signal)for 90 seconds than measure CO concentration ;
3) Go to step 1.

The MQ-7 sensor calibration experiment starts by performing heating and cooling cycles for 48 hours at a time before being calibrated. This step is necessary to be able to use it for a long period. In general, calibration of the MQ-7 sensor is performed in clean air by monitoring (reading) the output of the sensor. After 10 hours of monitoring, 100 (ppm) of CO are added to the air using a syringe and if the reading does not change by 1 to 2 points, the sensor is stopped heating. We repeat this experiment for the second MQ-7 sensor. Also, the calibration experiments made it possible to synthesize the fuzzy membership functions used in the BI-CO solution.

C. The IoT-AQM algorithm

Intelligent automatic air quality monitoring algorithm can be presented in the following pseudo-code 1:

D. The BI-CO fuzzy solution

In this subsection, we present the BI-CO fuzzy solution, where it estimates the AQI of the CO concentration by using a type-1 fuzzy inference system and two MQ-7 sensors. In which, six input membership functions are used to encode the CO concentration for each sensor as presented in figure 3 and figure 4, corresponding to very low **VL**, low **L**, moderate **M**, high **H**, very high **VH**, and ultra high **UH** CO concentration. They have allowed us to better define the limits of each concentration level through calibration experiments. For example, the low concentration level for the first sensor starts from 7 (ppm) (particle per million) to 14 (ppm) as presented in figure 3 in one hand. On the

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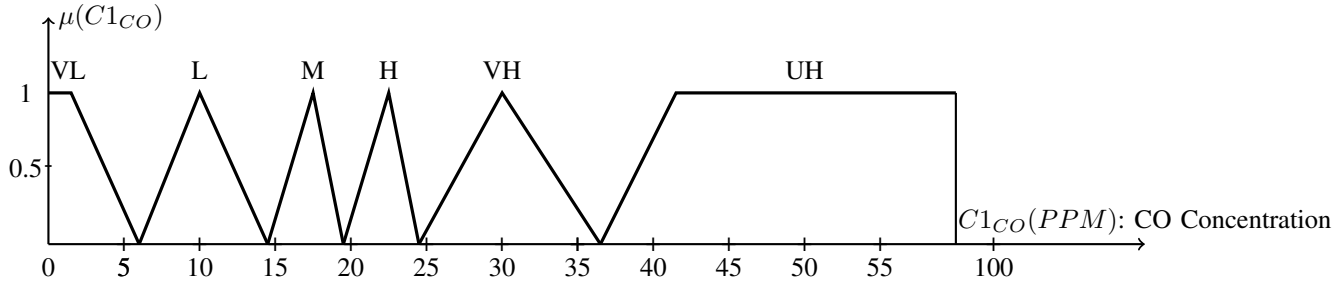

Fig. 3: Inputs membership functions of the first MQ-7 sensor

Algorithm 1 IoT-AQM algorithm

Execute the calibration procedure of the two MQ-7 sensors;

procedure IoT-AQM
while True **do**
1: Read the CO concentration from the first MQ-7 sensor;
2: Read the CO concentration from the second MQ-7 sensor;
3: Read the temperature Tm and humidity Hm from DH22 sensor;
4: Read the PM concentration from Shenyei PPD42NS sensor;
5: Execute the BI-CO procedure based on fuzzy logic for the calculation of the air quality index ;
6: Connecting to an Internet access point;
7: **IF** the connection is established **THEN** ;
Send the values of temperature Tm, humidity Hm, CO concentration, PM concentration and the air quality index AQI to a Google Drive Sheets folder ;
Else go to step 6;
END IF
8 :Display of temperature Tm, humidity Hm, CO concentration, air quality index AQI on OLED display ;
9: **IF** AQI of CO $\geq$ the dangerous value **THEN**;
Alarm $\leftarrow$ 1;
Else
Alarm $\leftarrow$ 0;
END IF
end while
end procedure

other hand, in the case of the second sensor, it starts from 5 (ppm) to 13 (ppm). Besides, in Figure 5, we choose to use six output membership functions to describe the ranges of air quality index, corresponding to the table I that presents the six air quality conditions: good, moderate, unhealthy for sensitive people, unhealthy, very unhealthy and hazardous.

In this work, we have proposed a set of thirty-six fuzzy rules to estimate the AQI as presented in table II. Moreover, this fuzzy reasoning is developed to avoid any disastrous consequences in case of a measurement failure on one of the two MQ-7 sensors.

Air quality Index	Levels of Health Concern	CO Breakpoints (PPM)
When the AQI is in this range :	.. air quality conditions are:	When the Breakpoint is in this range:
0 to 50	Good	0-8
51 to 100	Moderate	9-15
101 to 150	Unhealthy for Sensitive Groups	16-24
151 to 200	Unhealthy	25-34
201 to 300	Very Unhealthy	35-39
301 to 500	Hazardous	40-100

TABLE I: The AQI six levels of health concern. [19]

	C1	VL	L	M	H	VH	UH
C2	VL	G	M	M	USP	U	VU
	L	M	M	M	USP	U	VU
	M	M	M	M	USP	U	VU
	H	USP	USP	USP	U	VU	HZ
	VH	U	U	U	VU	VU	HZ
	UH	VU	VU	VU	HZ	HZ	HZ

TABLE II: The BI-CO fuzzy rules

The interpretation of the first rule is as follows: if the CO concentration of the first sensor (CO(1)) is very low (VL) and the CO concentration of the second sensor (CO(2)) is very low (VL), these then imply that the AQI is good (G). However, if the CO concentration of the first sensor (CO(1)) is very low (VL) and the CO concentration of the second sensor (CO(2)) is ultra-high (UH), then implies that the AQI is very unhealthy (VU).

III. Results and Discussion

To examine the efficiency of the realized IoT-fuzzy-based AQM device, we have performed several experiment tests. Figure 6 represents the implementation schematic diagram of the IoT-fuzzy-based AQM device in a house (Smart Home). We used this device in several places in the house: in

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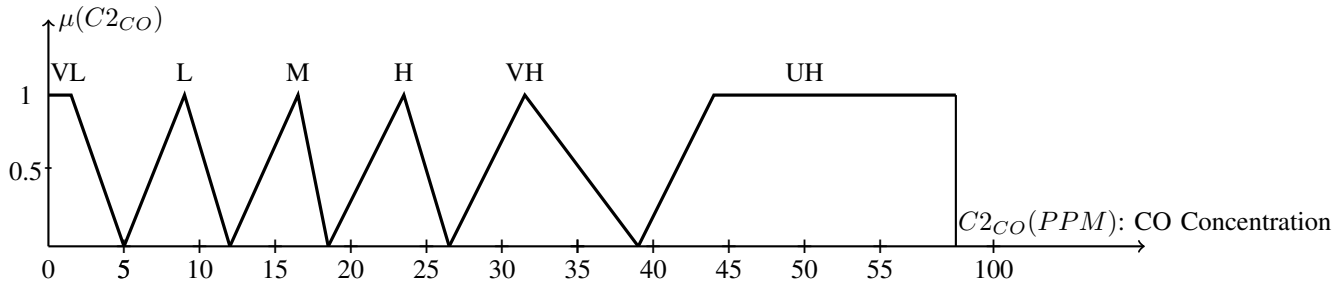

Fig. 4: Inputs membership functions of the second MQ-7 sensor

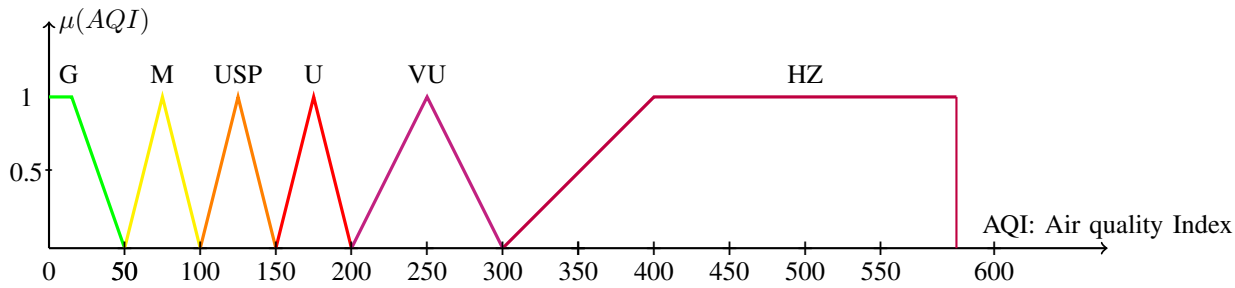

Fig. 5: Outputs membership functions of the Air quality Index

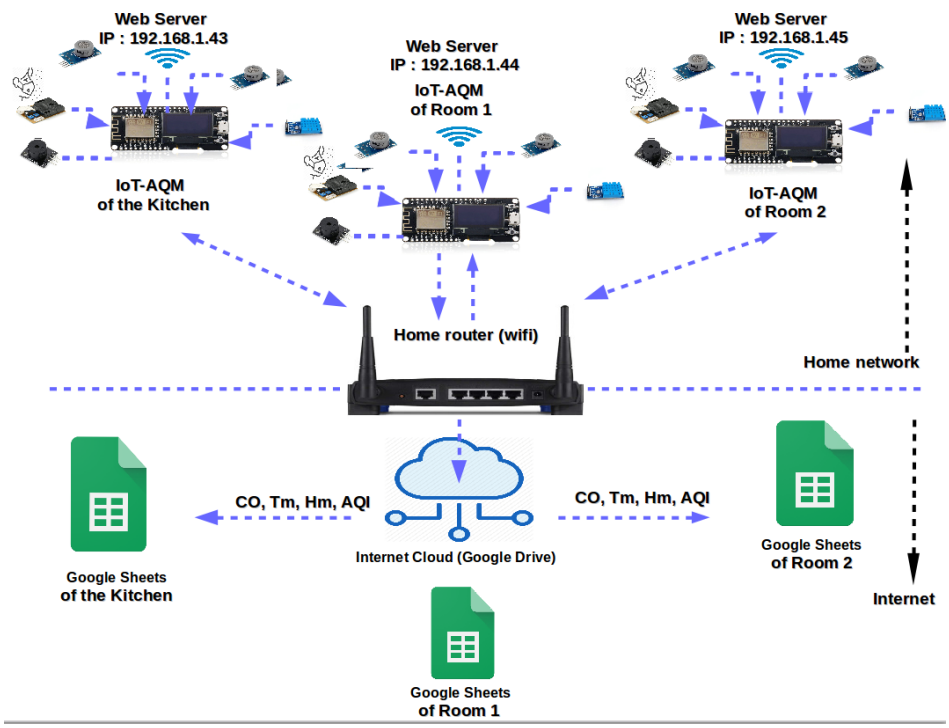

Fig. 6: Schematic diagram the air monitoring implementation in a Smart home

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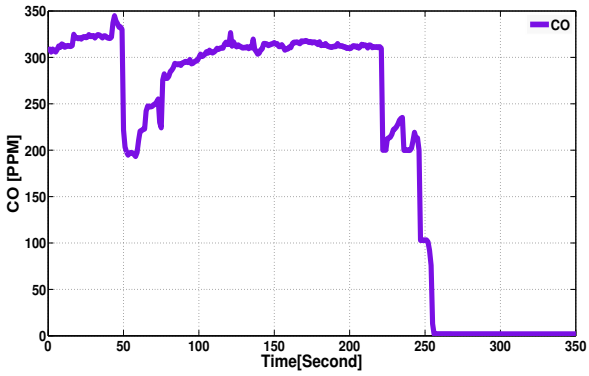

Fig. 7: Concentration graph of CO in the bathroom.

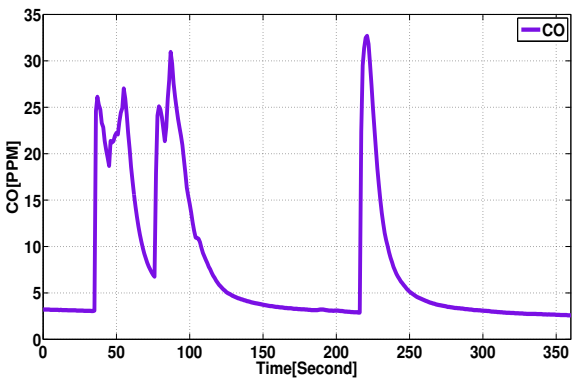

Fig. 8: Concentration graph of CO in the kitchen.

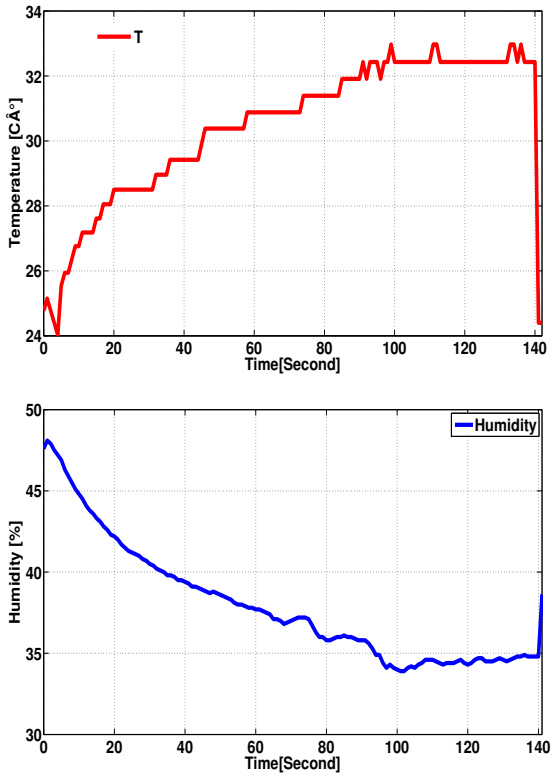

Fig. 9: Concentration graphs of the temperature and humidity in the kitchen.

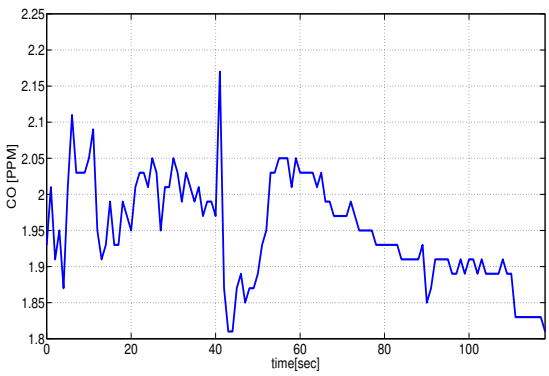

Fig. 10: Concentration graph of CO in room 1.

two rooms, in the kitchen, and the bathroom. Each device contains a web server with a well-defined IP address (ID) and they are connected to access points (Home router, 4G modem, Smart-phone, etc.). We have used the Google Drive service to make a web application under Google Sheets to collect in real-time the daily CO, temperature, humidity, and particulate matter concentrations.

a) : The first test was carried out in the bathroom. Figure 7 shows the CO concentration measured in the bathroom next to the water heater. The purpose of this test is to analyze CO concentration during incomplete combustion (the absence of oxygen). We notice that the system detected an average CO concentration equals 250 (ppm) at the moment 250 seconds (4 minutes), this value indicates that the air quality in the bathroom is dangerous. Afterward, we notice a progressive decrease of the CO value from the instant 260 seconds. Because of the aeration, we produced after hearing the alarm.

b) : The second test was carried out in the kitchen, in which figure 8 shows the CO concentration variation. We notice the average concentration equals 10 (ppm). According to table I the air quality in the kitchen is good. Figure 9 shows the recorded temperature and humidity concentrations in the kitchen. We notice that the temperature value generally increases although it goes down due to the opening of the kitchen door. Also, we notice that the temperature value generally decreases during this experience.

c) : The third test was carried out in room 1. Figure 10 shows the CO concentration measured in this room. We notice the average concentration equals 2.5 (ppm). According to table I the air quality is good in tis room. Figure 11 presents the PM concentration during the third test in room 2 , where it reaches 25 ($\mu g/m^3$) and the average concentration equals 5 (ppm). According to table I the air quality is good. We note that the use of this system has allowed us to analyze the air quality in all the house with an economic cost and a good accuracy. this solution can be generalized in many indoor environments such as factories, schools, hospitals, etc.

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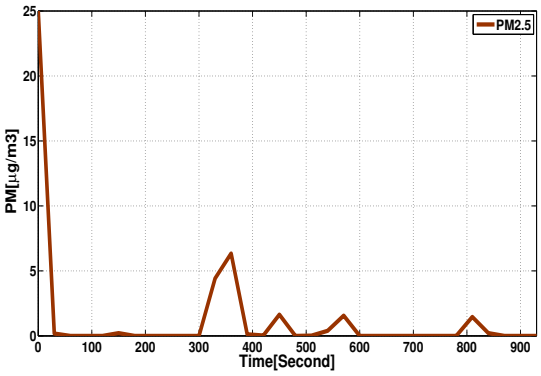

Fig. 11: PM concentration in room 2.

IV. Conclusion

The works presented in this paper are in the context of air quality monitoring in smart homes. This domain provides the citizen with an intelligent service on air quality in their living areas. For this purpose, we discussed the problem of automatic air quality monitoring in residential areas (home, hospital, factory, etc) using the IoT-fuzzy-based AQM device. In the first stage of this work, we have summarized a state of art on the subject of the use of the Internet of things for automatic air quality monitoring in the last three years. Then, we have present a low-cost system for pollutant measures such as carbon monoxide CO and particulate matter (PM). Afterward, we have present a calibrating procedure for the MQ-7 sensor. In the second step, we have proposed a solution based on fuzzy logic for the estimation of the air quality index named fuzzy BI-CO. This solution ensures the diagnosis of any faults that may occur on the MQ-7 sensor on one hand, and on the other hand, it enhances its accuracy. In the third step, we proposed an architecture for the implementation of the IoT-fuzzy-based AQM devices in a smart home. In general, the results of experiments tests have validated the effectiveness of using the IoT technology for air quality monitoring. In future works, we plan to use this system to estimate the air quality model based on machine learning techniques and the data recorded by the IoT-fuzzy-based AQM device for a year. Also, we will use the type-2 Fuzzy sets to enhance the BI-CO solution.

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